Transferability Reinforcement of Adversarial Attacks for Remote Sensing Image Classification via Hierarchical Transformation Composition

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***Abstract*—Adversarial attacks constitute an effective means of evaluating model robustness and revealing intrinsic weaknesses. Since practical model deployment typically adheres to black-box settings, existing attack methods often employ multiple input transformations to enhance the transferability of adversarial examples. However, remote sensing images often exhibit ambiguous foreground–background distinctions and various geospatial discrepancies, resulting in high model specificity in discriminative cues for classification. Consequently, the input patterns simulated through fixed transformation schemes are insufficient to prevent overfitting to the source model, thereby limiting the transferability of the generated adversarial examples. To solve this issue, we propose a hierarchical transformation composition (HTC) framework that reinforces adversarial transferability through a coordinated action execution (CAE) strategy. Specifically, the determination of input transformations is formulated as an adaptive action selection procedure, which is progressively executed by multilevel policy networks. Then, the policy networks are iteratively updated via proximal policy optimization (PPO) based on the advantage estimates from a shared value network. Moreover, a hybrid reward mechanism (HRM) is introduced to dynamically integrate loss information from both feature and output layers. Rather than directly imposing directional constraints on gradient calculation, the supervision is shifted to the optimization of policy networks, which prevents the sacrifice of intrinsic attack capacity while enhancing transferability. Extensive experiments on the UCM and SIRI-WHU datasets demonstrate that the proposed method achieves state-of-the-art performance across various model architectures. The code will be released at https://github.com/fuyimin96/HTC upon acceptance.**

***Index Terms*—Adversarial attack, hierarchical transformation composition (HTC), model robustness, remote sensing.**

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I. Introduction

REMOTE sensing image classification [1], [2], [3] plays a crucial role in a wide range of applications across both military and civilian domains. Despite remarkable performance improvements enabled by deep neural networks (DNNs) [4], [5], [6], [7], [8], their vulnerability to adversarial perturbations has raised serious concerns regarding model reliability. Consequently, the practical deployment of DNN-based remote sensing image classification methods remains constrained, particularly in safety-critical scenarios. Against this backdrop, adversarial attacks [9], [10] have emerged as an effective means to ensure reliable deployment by assessing model robustness and revealing intrinsic weaknesses beforehand.

Existing adversarial attack research primarily falls into two task scenarios: white-box attacks [11], [12], [13] and black-box attacks [14], [15], [16]. In the white-box setting, the internal architecture and parameter information of the target model are fully accessible to the attacker, making the manipulation of the model easy to accomplish. In contrast, model deployment in real-world applications typically adheres to the black-box setting. Since the specifications of the target model are inaccessible, the attack efficacy of adversarial examples generated on a surrogate source model should be transferable to unseen target models [17]. To fulfill this requirement, various techniques have been developed to enhance the transferability of adversarial examples, including gradient augmentation [18], [19], [20], feature disruption [21], [22], and input transformation [23], [24], [25]. Compared with other types of attack methods, input transformation-based methods offer distinct advantages in flexibility and extensibility. During the generation of adversarial examples, overfitting to model-specific information is effectively mitigated by diversifying input patterns in gradient calculation.

In parallel with the continuous progress in general computer vision, research on adversarial attacks has also gained increasing attention in the field of remote sensing [26], [27]. Although extensive explorations have been conducted across various sensing modalities and tasks [28], [29], [30], [31], [32], the intrinsic complexities of remote sensing images still pose severe challenges to the cross-model transfer of attack efficacy. Compared with natural images, remote sensing images often exhibit ambiguous foreground–background separation,

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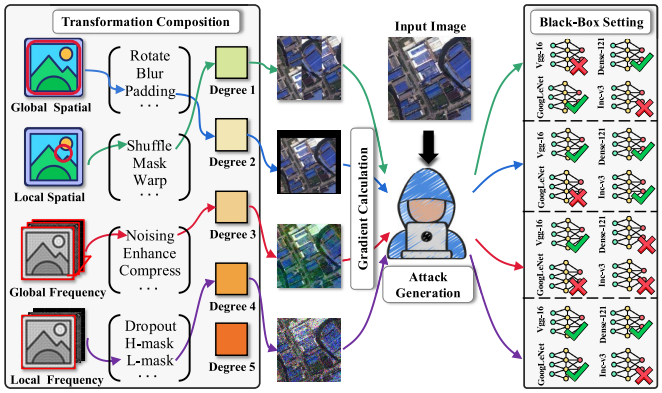


Fig. 1. Black-box attack effectiveness of adversarial examples generated with different transformation compositions using ResNet-34 as the source model.

pronounced geospatial discrepancies, and significant resolution variations [33]. Due to the lack of clear foreground delineation, their interpretation relies more on holistic scene-level contextual information rather than distinctive object-level patterns. This leads to substantial divergences in the discriminative cues exploited by different model architectures. Such intermodel discrepancies are further amplified by geospatial heterogeneity and resolution variations, making the adversarial example generation process more prone to overfitting to model-specific information. Under this scenario, achieving transferable adversarial attacks requires more diverse simulation of input patterns during gradient calculation. However, existing attack methods typically employ a fixed transformation scheme to simulate input patterns, which are insufficient to capture a generalized gradient landscape that reflects the shared vulnerabilities across different models. As a result, the transferability of the generated adversarial attacks for remote sensing image classification is limited. Moving beyond the refinement of standalone transformations, recent studies [34], [35] have demonstrated the potential of integrating multiple transformation strategies to enrich input diversity, thereby further enhancing adversarial transferability. Nevertheless, the enriched diversity is restricted to the combination or sequencing of predefined transformations, which themselves still remain fixed within a limited scope. As illustrated in Fig. 1, input transformations can be decomposed into factors specifying *where*, *what*, and *how* to transform. The hierarchical composition of these factors substantially expands the transformation space into a higher-dimensional decision space, where the generated adversarial examples exhibit heterogeneous perturbation coverage over the discriminative cues of different model architectures. Therefore, transitioning the transformation policy from sequential combination to hierarchical composition enables further diversification of potential input patterns. Within this expanded decision space, more generalizable perturbation updates can be achieved by adaptively adjusting the transformation strategy in response to the evolving gradient landscape. Accordingly, the challenges to adversarial transferability posed by remote sensing image characteristics can be effectively addressed.

Motivated by this insight, we propose a hierarchical transformation composition (HTC) framework to reinforce the adversarial transferability for remote sensing image classification. Different from existing attack methods, HTC reformulates the determination of input transformations as an adaptive action selection procedure. Specifically, a coordinated action execution (CAE) strategy is employed to determine hierarchical transformation factors during the iterative perturbation-generation process, with the decisions progressively executed by multilevel policy networks. Then, the policy networks are updated through proximal policy optimization (PPO) [36], driven by advantage estimates obtained from a shared value network, in order to better accommodate the evolving gradient landscape. Moreover, loss information from feature layers is dynamically integrated with that from the output layer through a hybrid reward mechanism (HRM), thereby steering the crafted perturbations toward stronger cross-model generalizability. Notably, this hybrid reward signal does not directly impose directional constraints on the gradient calculation for adversarial example generation but is instead shifted to the optimization of the policy networks, which prevents the sacrifice of intrinsic attack capacity while enhancing transferability. Finally, the transformed input variants are jointly used for gradient calculation. Extensive experiments on the UCM [37] and SIRI-WHU [38] datasets demonstrate that the proposed method consistently achieves state-of-the-art attack performance across diverse model architectures in both white- and black-box settings.

Our main contribution can be summarized as follows.

1) We propose HTC, which determines the hierarchical composition of input transformations through adaptive action selection, thereby reinforcing adversarial transferability for remote sensing image classification.
2) We develop a CAE strategy that progressively determines transformation factors through iteratively updated multilevel policy networks to better accommodate the evolving gradient landscape.
3) We design an HRM to integrate loss information from the feature and output layers. By shifting the supervision from gradient manipulation to policy-network optimization, the transferability is further enhanced without sacrificing intrinsic attack capacity.

The remainder of this article is organized as follows. Section II reviews the related work. Section III introduces the problem formulation and describes the technical details of the proposed method. Section IV reports the experimental results and corresponding analyses. Finally, Section V concludes this article and outlines potential directions for future research.

II. Related Works

In this section, we first provide a brief review of related works on adversarial attacks and defenses in the general computer vision field. Then, we introduce the ongoing advances of adversarial machine learning in the field of remote sensing.

A. Adversarial Attacks

From the perspective of task scenarios, existing adversarial attack methods can be broadly grouped into two categories: white-box attacks and black-box attacks.

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Early white-box attack methods assume full access to model architecture and parameters, enabling deliberately crafted adversarial examples to achieve high attack success rates. Based on loss-gradient information, Szegedy et al. [11] employed a box-constrained optimization strategy to generate imperceptible perturbations that substantially amplify prediction errors. However, the substantial computational overhead introduced by repeated line searches severely limits its practicality in the real world. To overcome this limitation, Goodfellow et al. [12] proposed the fast gradient sign method (FGSM) to realize efficient adversarial example generation via a single-step update. Kurakin et al. [13] then extended FGSM into an iterative formulation, I-FGSM, which boosts attack strength through multistep gradient refinement. Moreover, Madry et al. [39] introduced projected gradient descent (PGD) to constrain the iteratively updated perturbation within the ℓ_∞-ball. As a special case, Kurakin et al. [40] incorporated target-class information into the loss function to force the model toward a specified misclassification.

Since internal specifications of target models are inaccessible in black-box settings, some initial attempts tried to estimate their gradients by repeatedly querying model outputs [16], [41]. However, the excessive query cost rendered them highly inefficient in real applications. Therefore, various techniques have been developed to improve the transferability of adversarial examples crafted on the surrogate source model. To alleviate overfitting to local maxima of the loss landscape, various gradient augmentation operations have been incorporated into iterative adversarial generation, thereby stabilizing the gradient update. Dong et al. [14] integrated a momentum term into the optimization process to enhance the generalizability of adversarial perturbations across different model architectures. Building upon this, Lin et al. [19] leveraged the looking-ahead property of Nesterov's accelerated gradient to obtain a more transferable update direction. Wang et al. [18] further enhanced the momentum accumulation process by aggregating gradients from multiple data points. To achieve a broader attack scope, Zhu et al. [20] leveraged gradient relevance from neighboring data points, while Wang et al. [42] introduced a global momentum initialization approach to approximate the global optimum. Differently, feature disruption-based methods primarily focus on improving adversarial transferability by selectively corrupting the critical features for decision-making. Wang et al. [21] manipulated features with high importance to object awareness to generate transferable adversarial examples. For more precise importance estimation, Zhang et al. [22] leveraged neuron attribution to achieve computation-efficient transferability improvement. Huang et al. [43] devised a fine-tuning scheme for existing attacks by enlarging the perturbations on intermediate representations. Beyond the two types of methods above, input transformation-based methods draw on the idea of data augmentation and mitigate entrapment in source-model local optima by simulating diverse input patterns. Xie et al. [23] pioneered this line of research by randomly incorporating preprocessing operations into the attack process. Subsequently, a series of input-transformation strategies were proposed, which leverage translation invariance [24], scale invariance [19], semantic mixing [44], detail enrichment [45], and posteriori estimation [46] to further enhance the efficiency of black-box attacks. Moreover, several works have also explored transformations in the frequency domain [25] and the shuffled block spaces [47] to improve adversarial transferability. Apart from developing specific transformation strategies, Yuan et al. [34] explored the potential of multiple transformation combinations to boost attack performance. Recently, Zhu et al. [35] designed a trajectory search framework to learn the optimal transformation sequence via reinforcement learning (RL).

B. Adversarial Defenses

Given the vulnerabilities exposed by adversarial attacks, various adversarial defense methods [48] have been developed to improve model robustness, which are mainly achieved through two routes: adversarial training (AT) and input reconstruction.

As a conceptually intuitive solution, AT [12], [49] improves model robustness by integrating adversarially perturbed inputs into model training. From the perspective of min–max optimization, Madry et al. [39] strengthened resistance to perturbations by minimizing the worst case loss. Zhang et al. [50] designed a regularized surrogate loss to achieve a theoretically grounded tradeoff between natural and boundary errors. Tramer et al. [51] introduced ensemble AT, which enhances adversarial robustness by augmenting training data with perturbations crafted from multiple models. For foundation models, tailored AT strategies have also been implemented, such as continuous adversarial preference optimization [52] and learnable text-prompt tuning [53].

Unlike AT, input reconstruction-based methods aim to recover the underlying representation from perturbed inputs, thereby ensuring the fidelity of input signals for model inference. Liao et al. [54] reduced input perturbations by performing denoising under the guidance of high-level feature representations. Guo et al. [55] utilized JPEG compression to remove perturbations from input images. Without degrading classification accuracy, Liu et al. [56] redesigned a DNN-oriented JPEG compression framework through feature distillation to rectify adversarial examples. Yang et al. [57] disentangled class-essential and class-redundant information by a variational autoencoder to detect and defend against adversarial attacks. Considering generalizable defense against unknown attacks, Zhang et al. [58] proposed a metainvariance defense approach to extract attack-invariant representations.

C. Adversarial Machine Learning in Remote Sensing

Driven by the growing need for reliable model deployment, adversarial machine learning has also made continuous progress in the field of remote sensing, prompting an ongoing interplay between attacks and defenses.

Xu et al. [26] took an early step by systematically examining the vulnerability of DNNs for remote sensing scene classification. Inspired by the high similarity of shallow-layer representations, Xu et al. [27] proposed a feature-level mixup attack to mislead multiple models. Fu et al. [59] introduced a spatial-frequency cotransformation (SFCoT) strategy to enrich target-related input variations for stronger black-box

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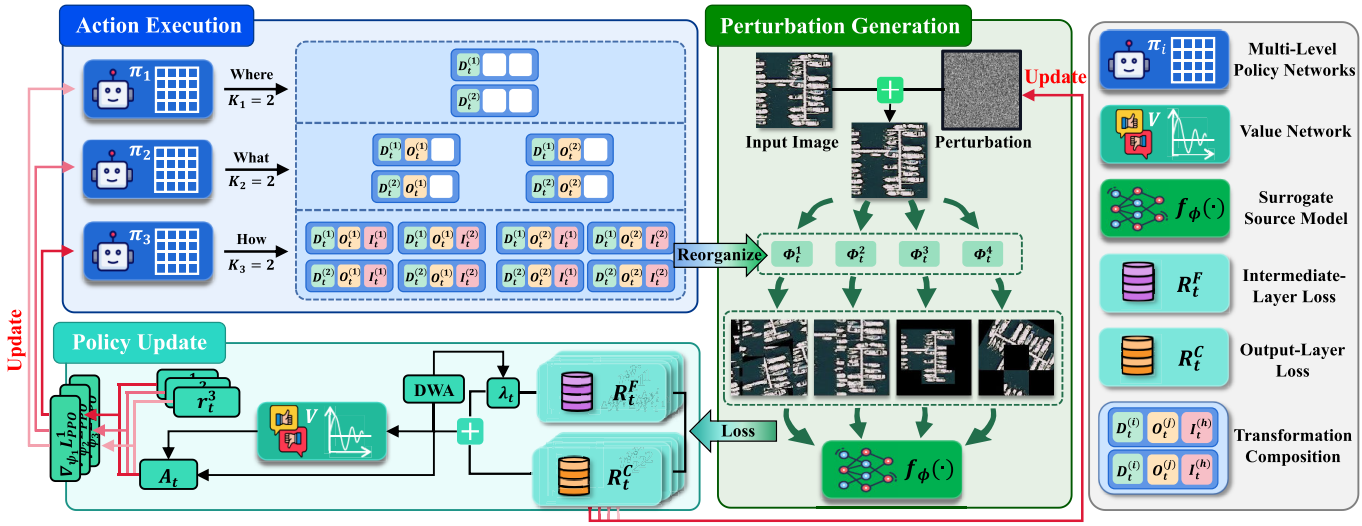


Fig. 2. Overview of the pipeline for generating adversarial examples using the proposed HTC framework. The procedure involves three key stages: CAE, iterative perturbation generation, and dynamic policy update.

attack efficacy. Dang et al. [60] proposed a plug-and-play amplitude normalization (AMPN) technique to diversify the frequency characteristics of remote sensing images. Liu et al. [61] devised asymmetric frequency mixing (AFM) to further mitigate intermodel discrepancies. Taking the characteristics of remote sensing images into account, Luo et al. [62] proposed a style transfer-enabled adversarial attack for scene-level perturbation generation. On the defense side, Su et al. [63] adopted a reconstruction-assisted AT pipeline to improve model robustness. Yu et al. [64] leveraged pretrained diffusion models to reduce the susceptibility to adversarial examples. Li et al. [65] proposed a robust feature augmentation strategy to ensure reliable object discrimination. Beyond classification, adversarial machine learning has also been explored in other remote sensing tasks, such as object detection [31], [66], semantic segmentation [32], and image retrieval [67].

In addition to optical images, Xu et al. [28] investigated the impact of adversarial attacks on hyperspectral image (HSI) classification. Correspondingly, Zhang et al. [68] proposed a multiscale state space model to ensure reliable HSI classification under input distortions. Moving beyond single-modality settings, Chen et al. [69] conducted a more comprehensive analysis of the impacts of diverse adversarial examples across different remote sensing data modalities. Hu et al. [30] advanced cooperative adversarial generation for multimodal remote sensing data through a multiloss optimization paradigm. Besides image representations, the electromagnetic characteristics of synthetic aperture radar (SAR) data have also been leveraged for both the attack [29] and defense [70] of automatic target recognition systems.

III. Methodology

Fig. 2 illustrates the adversarial example generation process of the proposed HTC framework for remote sensing image classification. During gradient calculation, the applied transformations are expressed as the composition of hierarchical factors. The corresponding selection actions are then progressively executed by multilevel policy networks to simulate diverse input patterns. Given the transformed image set, adversarial examples are iteratively updated using the aggregated gradient calculated on a surrogate model. Meanwhile, to adapt to the evolving gradient landscape, the policy networks are simultaneously optimized under the guidance of a shared value network, with advantages estimated through an HRM. In the remainder of this section, we first introduce the reformulated problem definition of input transformation-based adversarial attacks, followed by the technical details of how the proposed HTC framework reinforces adversarial transferability.

A. Problem Definition

The adversarial attack task is commonly formulated as a constrained optimization problem. For a clean input image x with ground-truth label y, the goal is to fool the target model $f_\theta(\cdot)$ by the generated perturbed image $\tilde{x}$ with an imperceptible perturbation

$$\underset{\|\tilde{x}-x\|_\infty \le \epsilon}{\operatorname{argmax}} J(\tilde{x}, y; \theta) \tag{1}$$

where ϵ denotes the radius of the ℓ_∞-norm ball constraint for perturbation magnitude and $J(\cdot)$ is the classification loss function. Since the parameters θ of the target model are inaccessible under black-box settings, the iterative update of $\tilde{x}$ relies on gradients obtained from a surrogate source model $f_\phi(\cdot)$. At the tth iteration, the update of the perturbed image is performed with step size α

$$\tilde{x}_t = \operatorname{Clip}_{x,\epsilon}\left\{\tilde{x}_{t-1} + \alpha \cdot \operatorname{sign}\left(\nabla_{\tilde{x}_{t-1}} J\left(\mathcal{T}\left(\tilde{x}_{t-1}\right), y; \phi\right)\right)\right\} \tag{2}$$

where $\mathcal{T}(\cdot)$ and $\operatorname{Clip}_{x,\epsilon}\{\cdot\}$ refer to the applied input transformation and pixelwise clipping operation, respectively.

To further enhance adversarial transferability, input transformation-based attacks typically calculate the average gradient over a set of N transformed images produced by

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TABLE I

ILLUSTRATION OF SOME RESULTING TRANSFORMATIONS DERIVED FROM COMPOSING HIERARCHICAL FACTORS. FOR EACH DOMAIN AND OPERATION TYPE, THE IMAGES WITHIN A CELL DEPICT VARIANTS OF DIFFERENT INTENSITIES

Domain	Global		Local	
Spatial	Rotate	Padding	Shuffle	Mask
				
Frequency	Smooth	Noising	Dropout	L-mask
				

an ensemble of transformation operations, i.e., $\Psi = \{\mathcal{T}_n\}_{n=1}^{N}$. Instead of selecting fixed transformations from a predefined scope, we reformulate the determination of applied transformations as the composition of L hierarchical transformation factors

$$\Psi = \{\mathcal{T}_n\} = \left\{\left(\mathcal{F}_n^1 \oplus \cdots \oplus \mathcal{F}_n^L\right)\right\}_{n=1}^{N} \tag{3}$$

where $\oplus$ represents the composition operation for factors $\{\mathcal{F}^l\}_{l=1}^{L}$ across different hierarchies. This reformulation expands the transformation decision space from N transformation-level selections to $N \times L$ factor-level determinations, thereby enabling more diverse simulation of input patterns. Accordingly, the objective of adversarial attacks can be expressed as finding the optimal transformation composition Ψ^* that maximizes the expectation in the following:

$$\Psi^* = \underset{\Psi}{\operatorname{argmax}} \left(\mathbb{E}\left[J\left(\tilde{x}_\phi^\Psi, y; \theta\right)\right]\right) \tag{4}$$

where $\tilde{x}_\phi^\Psi$ denotes the adversarial example generated by the source model under the composed transformation ensemble Ψ.

B. CAE Strategy

The construction of the transformation ensemble for gradient calculation typically follows two paradigms. As a common choice, multiple transformations are integrated according to predefined schemes derived from analytical principles. In comparison, some methods aim to determine transformation policies in a learnable manner to better capture transferable gradient cues. However, existing learning policies remain confined to combining or sequencing fixed transformations within a given scope, which fails to properly coordinate the interactions required for hierarchical factor composition.

To overcome this limitation, we develop a CAE strategy to progressively determine input transformations. Specifically, the transformation composition is instantiated in three hierarchies ($L = 3$), corresponding to factor decisions of *where*, *what*, and *how* a transformation is applied. Some examples of the composed transformations are illustrated in Table I. Then, the execution of these decisions is carried out through multilevel policy networks $\Pi = \{\pi_1, \pi_2, \pi_3\}$, with a depth of three that is consistent with the hierarchies of the transformation composition, where each policy selects the factor at its corresponding hierarchy. The structured action at iteration t is denoted by $a_t = \{D_t, O_t, I_t\}$, where D_t, O_t, and I_t denote the decisions for selecting transformation factors at each hierarchy, respectively. The state s_t is constructed from the current perturbed input $\tilde{x}_{t-1}$. With each decision conditioned on decisions made in the preceding levels, the coherent action selection process can be modeled as the following conditional probability distributions:

$$\Pi\left(a_t \mid s_t\right) = \begin{cases} \pi_1\left(D_t \mid s_t\right) = P\left(D_t \mid \tilde{x}_{t-1}; \psi_1\right) \\ \pi_2\left(O_t \mid s_t, D_t\right) = P\left(O_t \mid \tilde{x}_{t-1}, D_t; \psi_2\right) \\ \pi_3\left(I_t \mid s_t, D_t, O_t\right) = P\left(I_t \mid \tilde{x}_{t-1}, D_t, O_t; \psi_3\right) \end{cases} \tag{5}$$

where ψ_l denotes the learnable parameters of the policy network at each level. For each hierarchy l, the corresponding policy π_l governs the selection of actions over a candidate set of size M_l. Accordingly, the policy networks are composed of parameter groups $\{\psi_1, \psi_2, \psi_3\}$ with sizes M_1, $M_1 \times M_2$, and $M_1 \times M_2 \times M_3$, which define conditionally normalized distributions as

$$\begin{cases} \sum_{m=1}^{M_1} \pi_1\left(D_t^{(m)} \mid s_t\right) = 1 \\ \sum_{n=1}^{M_2} \pi_2\left(O_t^{(n)} \mid s_t, D_t\right) = 1 \\ \sum_{p=1}^{M_3} \pi_3\left(I_t^{(p)} \mid s_t, D_t, O_t\right) = 1. \end{cases} \tag{6}$$

To construct the transformation ensemble, we execute multiple actions from each hierarchy

$$\begin{aligned} \left\{D_t^{(i)}\right\}_{i=1}^{K_1} &\sim \pi_1\left(\cdot \mid s_t\right) \\ \left\{O_t^{(j)}\right\}_{j=1}^{K_2} &\sim \pi_2\left(\cdot \mid s_t, D_t^{(i)}\right) \\ \left\{I_t^{(h)}\right\}_{h=1}^{K_3} &\sim \pi_3\left(\cdot \mid s_t, D_t^{(i)}, O_t^{(j)}\right) \end{aligned} \tag{7}$$

where K_1, K_2, and K_3 denote the execution multiplicities. Finally, the transformation ensemble can be formulated as

$$\Psi_t = \left\{\mathcal{T}_t^{(i,j,h)}\right\} = \left\{\left(\mathcal{F}^1\left(D_t^{(i)}\right) \oplus \mathcal{F}^2\left(O_t^{(j)}\right) \oplus \mathcal{F}^3\left(I_t^{(h)}\right)\right\} \tag{8}$$

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where each index (i, j, h) corresponds to a unique triplet ranging over $i \in [1, K_1]$, $j \in [1, K_2]$, $h \in [1, K_3]$. Therefore, the objective of adversarial attack is to learn the optimal hierarchical policies $\Pi^* = \{\pi_1^*, \pi_2^*, \pi_3^*\}$ such that the induced transformation decisions at each iteration achieve

$$\Pi^* = \underset{\{\pi_1,\pi_2,\pi_3\}}{\operatorname{argmax}} \mathbb{E}_{a_t \sim P(a_t|s_t)} \left[J\left(\tilde{x}_\phi^{\Psi_t}, y\right)\right] \tag{9}$$

where $P(a_t \mid s_t)$ denotes the joint action distribution induced by the overall transformation composition policy

$$P(a_t \mid s_t) = \pi_1(D_t \mid s_t)\, \pi_2(O_t \mid s_t, D_t)\, \pi_3(I_t \mid s_t, D_t, O_t). \tag{10}$$

C. Proximal Actor–Critic Update of Policy Networks

Once the coordinated determination of transformation factors is cast into a learnable action execution procedure, the improvement of adversarial transferability hinges on whether the multilevel policy networks can effectively adapt to the evolving gradient landscape during perturbation generation. Existing methods typically update the policies using losses computed on surrogate source models. However, the attack loss on the source model cannot directly assess the contribution of the employed transformations to adversarial transferability, rendering such supervision signals inherently unreliable.

As a remedy, we update the multilevel policy networks in an actor–critic manner using PPO. In the context of RL involving hierarchical decision processes, PPO is a widely adopted policy-gradient algorithm for optimizing parameterized policies. Based on advantage estimates from a value function (critic), the policy network (actor) is updated by maximizing a clipped surrogate objective, enabling stable improvement in the expected cumulative return. Given the constructed transformation ensemble Ψ_t, the classification loss on the surrogate source model is taken as the action value

$$R_t^C = J(\Psi_t(\tilde{x}_{t-1}), y; \phi). \tag{11}$$

Then, we employ a unified value network, which serves as the critic by estimating the expected return of the current state under the joint action distribution

$$V(s_t) = \mathbb{E}_{a_t \sim P(a_t|s_t)} \left[J\left(\Psi_t^{a_t}(\tilde{x}_{t-1}), y; \phi\right)\right]. \tag{12}$$

Building upon this, the advantage of the coordinated actions is computed as

$$A_t = R_t^C - V(s_t) \tag{13}$$

where a larger advantage indicates a better-than-expected contribution of the sampled transformation composition. Subsequently, the policy update rule at each hierarchy is formulated using the clipped surrogate objective

$$\mathcal{L}_{\text{PPO}}^l = \mathbb{E}\left[\min\left(r_t^l A_t,\ \text{clip}\left(r_t^l,\ 1-\xi,\ 1+\xi\right) A_t\right)\right] \tag{14}$$

where ξ denotes the clipping range and the probability ratios r_t^l are defined as

$$r_t^1 = \frac{\pi_1(D_t \mid s_t; \psi_1)}{\pi_1\left(D_t \mid s_t; \psi_1^{\text{old}}\right)}$$

$$r_t^2 = \frac{\pi_2(O_t \mid s_t, D_t; \psi_2)}{\pi_2\left(O_t \mid s_t, D_t; \psi_2^{\text{old}}\right)}$$

$$r_t^3 = \frac{\pi_3(I_t \mid s_t, D_t, O_t; \psi_3)}{\pi_3\left(I_t \mid s_t, D_t, O_t; \psi_3^{\text{old}}\right)} \tag{15}$$

where ψ_l^{old} represents the parameters of policy networks from the previous iteration. Finally, the parameters of policy networks at each hierarchy are optimized via gradient ascent

$$\psi_l \leftarrow \psi_l + \eta \nabla_{\psi_l} \mathcal{L}_{\text{PPO}}^l, \quad l \in \{1, 2, 3\} \tag{16}$$

where η is the learning rate.

D. Hybrid Reward Mechanism

In addition to the loss information from the output layer (i.e., the classification loss), some methods [43], [71] incorporate auxiliary supervision from intermediate feature layers to further improve adversarial transferability. However, such gains in transferability often come at the cost of intrinsic attack capacity, as the resulting perturbation updates tend to deviate from the direction for maximizing misclassification.

Taking the intermediate level attack (ILA) [43] as an example, when the update of adversarial example $\tilde{x}$ is oriented to maximize the output-layer loss, the gradient is given by

$$g = \nabla_{\tilde{x}} J(\tilde{x}, y; \phi), \quad u = \frac{g}{\|g\|} \tag{17}$$

where u denotes the most effective direction for increasing attack strength on the source model. If an intermediate-layer loss $L(\cdot)$ is introduced to fine-tune $\tilde{x}$ into $\hat{x}$, the corresponding gradient becomes

$$h = \nabla_{\hat{x}} L(\tilde{x}, \hat{x}; \phi), \quad v = \frac{h}{\|h\|} \tag{18}$$

and the direction v generally deviates from u. Under these definitions, the misclassification risk after a single-step update can be approximated by the first-order expansion of the classification loss

$$J(\tilde{x} + \alpha u, y; \phi) \approx J(\tilde{x}, y; \phi) + \alpha \|g\|. \tag{19}$$

In contrast, the intermediate-layer refinement yields

$$J(\tilde{x} + \alpha v, y; \phi) \approx J(\tilde{x}, y; \phi) + \alpha \|g\| \cos\varphi \tag{20}$$

where φ is the angle between u and v. Therefore, directly imposing the intermediate-layer loss on the perturbation update inevitably undermines the intrinsic attack capacity.

To overcome this dilemma, we design an HRM to prevent the sacrifice of attack capacity when pursuing intermediate-layer-driven transferability enhancements. Rather than being directly applied for gradient manipulation, the integrated loss information from feature and output layers is repurposed as a hybrid reward signal that serves as the action value in (13) for policy optimization

$$R_t = R_t^C + \lambda_t R_t^F \tag{21}$$

where R_t^F denotes the representation discrepancy derived from intermediate-layer objectives. Inspired by dynamic weight averaging (DWA) [72], the weight coefficient λ_t, initialized as λ at the first iteration, is dynamically adjusted according to

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the relative dynamics of the two supervision components to stabilize the optimization

$$\lambda_t = \frac{\exp\left(\omega_t^F\right)}{\exp\left(\omega_t^C\right) + \exp\left(\omega_t^F\right)} \tag{22}$$

where

$$\omega_t^C = \frac{\mathbb{E}_{a_t \sim P(a_t|s_t)}\left[R_t^C\right]}{\mathbb{E}_{a_{t-1} \sim P(a_{t-1}|s_{t-1})}\left[R_{t-1}^C\right]}$$
$$\omega_t^F = \frac{\mathbb{E}_{a_t \sim P(a_t|s_t)}\left[R_t^F\right]}{\mathbb{E}_{a_{t-1} \sim P(a_{t-1}|s_{t-1})}\left[R_{t-1}^F\right]}. \tag{23}$$

E. HTC Attack Algorithm

Algorithm 1 HTC Attack Algorithm

Input: The input image x with ground-truth label y; The source model $f_\phi(\cdot)$ and multi-level policy networks Π; Action execution multiplicities K_1, K_2, and K_3; Perturbation constraint ϵ, number of iterations T, step size α, clipping range ξ, initial weight coefficient λ, and momentum coefficient μ.

Output: The adversarial example $\tilde{x}$.

Initialize: $g_0 = 0$, $\tilde{x}_0 = x$.
for $t = 1$ **to** T **do**
  Determine transformation factors by Eq. (7);
  Construct and reorganize the transformation ensemble $\Psi_t = \{\Phi_t^s\}_{s=1}^{K_2K_3}$ by Eqs. (8) and (25);
  for $s = 1$ **to** K_2K_3 **do**
    Get the transformed image $\Phi_t^s(\tilde{x}_{t-1})$;
  Adjust the weight coefficient λ_t by Eq. (22) and calculate the hybrid reward signal R_t by Eq. (21);
  Estimate the advantages by Eq. (13);
  Update multi-level policy networks based on the rule formulated in Eq. (14);
  Calculate the gradient $\bar{g}_t$ over transformed images by Eq. (26) and update momentum by Eq. (27);
  Update the generated adversarial example $\tilde{x}_t$ based on the gradient sign by Eq. (28);
end for
return $\tilde{x} = \tilde{x}_T$

After executing the coordinated action selection procedure described above, the transformation ensemble used for gradient calculation at iteration t is determined as

$$\Psi_t = \left\{\mathcal{T}_t^{(1,1,1)}, \ldots, \mathcal{T}_t^{(K_1,K_2,K_3)}\right\}. \tag{24}$$

Rather than applying these transformations independently to produce input variants, we reorganize the ensemble according to the first-hierarchy decisions $D_t^{(i)}$. Transformations that share the same second-/third-hierarchy factors (j,h) are grouped into K_2K_3 sequences

$$\Psi_t = \left\{\Phi_t^s\right\}_{s=1}^{K_2K_3} = \left\{\mathcal{T}_t^{(1,j,h)} \cup \cdots \cup \mathcal{T}_t^{(K_1,j,h)}\right\} \tag{25}$$

where $(j,h) \in [K_2] \times [K_3]$. This structured grouping further enriches the simulated input patterns, enabling the exploration of a more generalized gradient landscape that better captures discriminative cues across different model architectures. Following the common practice in existing black-box attack methods, the gradient is calculated across all transformation sequences

$$\bar{g}_t = \frac{1}{K_2K_3} \sum_{s=1}^{K_2K_3} \nabla_{\tilde{x}_{t-1}} J\left(\Phi_t^s\left(\tilde{x}_{t-1}\right), y, \phi\right) \tag{26}$$

and the update is then performed via

$$g_t = \mu g_{t-1} + \frac{\bar{g}_t}{\|\bar{g}_t\|_1} \tag{27}$$

where μ denotes the momentum coefficient. In this case, the generation of adversarial examples can be formulated as

$$\tilde{x}_t = \mathrm{Clip}_{x,\epsilon}\left\{\tilde{x}_{t-1} + \alpha \cdot \mathrm{sign}\left(g_t\right)\right\}. \tag{28}$$

The adversarial example generation process of HTC is outlined in Algorithm 1.

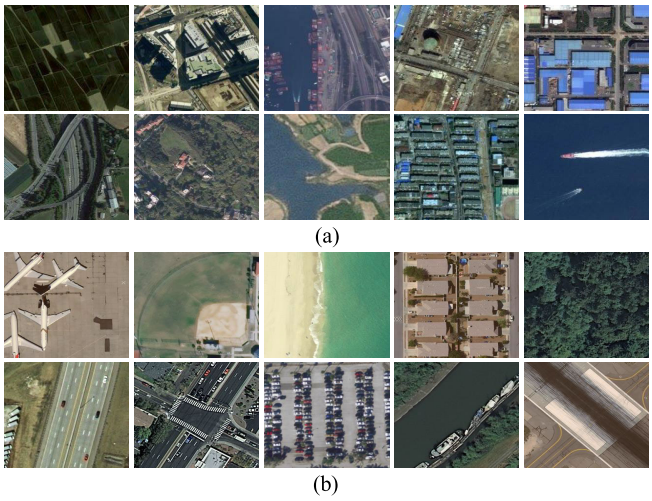


Fig. 3. Representative image samples from the datasets used in our experiments. (a) UCM and (b) SIRI-WHU.

IV. Experiments

A. Experimental Setup

1) Dataset: We conducted experiments on two widely used remote sensing image classification datasets, UCM [37] and SIRI-WHU [38], to evaluate the attack performance of different methods. These two datasets contain images of 21 and 12 land cover types, respectively, with representative samples illustrated in Fig. 3. In the experiments, all images are resized to 224×224.

2) Model Architectures: In our experiments, six commonly used DNN architectures are used to benchmark different adversarial attack methods: VGG16 (Vgg-16) [4], ResNet-34 (Res-34) [6], DenseNet-121 (Dense-121) [7], Inception-ResNet-V2 (IncRes-v2) [73], Inception-V3 (Inc-v3) [74], and GoogLeNet [5]. Each architecture is used in turn as the source model for generating adversarial examples, and the resulting attack performance is evaluated across all architectures.

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TABLE II

ATTACK SUCCESS RATES ON THE UCM DATASET AGAINST DIFFERENT MODELS. THE BEST ATTACK PERFORMANCE ON EACH MODEL IS BOLDED

Model	Attack	Vgg-16	Res-34	Dense-121	IncRes-v2	Inc-v3	GoogLeNet
Vgg-16	MIM	96.7%	37.7%	40.9%	40.5%	34.0%	38.6%
	DIM	**100.0%**	40.5%	48.6%	14.4%	41.9%	38.6%
	ILA	56.7%	11.6%	13.0%	46.7%	14.0%	20.7%
	SIM	98.1%	39.1%	47.9%	43.3%	43.3%	45.6%
	Admix	98.6%	45.1%	54.9%	41.9%	45.1%	54.0%
	SSA	97.2%	40.0%	45.6%	44.7%	45.1%	49.8%
	GIM	95.4%	37.7%	41.9%	42.8%	38.6%	41.9%
	BSR	96.7%	52.1%	56.3%	45.1%	50.7%	54.9%
	Mixcut	16.3%	8.4%	17.2%	**30.7%**	15.4%	14.4%
	NWSP	32.1%	10.2%	12.1%	20.9%	15.8%	7.9%
	SFCoT	99.1%	55.8%	61.4%	46.1%	59.1%	61.4%
	AMPN	**100.0%**	37.7%	40.0%	40.7%	42.3%	49.3%
	AFM	99.1%	61.3%	72.4%	41.8%	59.1%	65.8%
	L2T	97.7%	72.9%	84.4%	47.7%	70.2%	76.2%
	HTC	**100.0%**	**73.6%**	**87.1%**	**47.9%**	**79.1%**	**78.7%**
Res-34	MIM	34.4%	**100.0%**	33.5%	44.2%	26.5%	39.5%
	DIM	19.1%	**100.0%**	39.5%	5.1%	19.5%	29.8%
	ILA	18.1%	52.1%	15.8%	49.0%	13.0%	23.3%
	SIM	48.4%	**100.0%**	47.0%	41.4%	41.9%	54.9%
	Admix	57.2%	**100.0%**	55.4%	40.5%	45.1%	57.2%
	SSA	47.0%	99.5%	47.9%	47.9%	42.3%	55.5%
	GIM	34.0%	99.1%	33.5%	47.4%	30.2%	43.3%
	BSR	67.4%	97.7%	76.3%	45.1%	56.7%	74.0%
	Mixcut	22.8%	15.8%	24.7%	47.0%	22.3%	27.4%
	NWSP	12.6%	19.5%	7.4%	26.5%	16.3%	9.7%
	SFCoT	72.1%	97.2%	78.1%	46.5%	67.0%	75.5%
	AMPN	34.4%	**100.0%**	43.3%	42.3%	28.8%	42.3%
	AFM	83.1%	99.6%	88.5%	48.0%	77.8%	84.1%
	L2T	79.4%	98.6%	88.5%	49.1%	79.9%	86.2%
	HTC	**85.6%**	**100.0%**	**91.6%**	**50.2%**	**84.2%**	**90.2%**
Dense-121	MIM	38.6%	35.4%	99.5%	43.3%	28.4%	41.4%
	DIM	21.9%	34.4%	**100.0%**	5.1%	27.9%	30.2%
	ILA	40.8%	34.4%	72.4%	46.1%	28.4%	41.4%
	SIM	57.2%	51.6%	**100.0%**	40.0%	50.2%	59.1%
	Admix	62.8%	60.5%	**100.0%**	43.3%	60.5%	67.9%
	SSA	49.3%	47.4%	99.1%	44.7%	44.2%	53.5%
	GIM	35.8%	34.9%	98.6%	46.1%	23.7%	39.1%
	BSR	68.4%	63.3%	99.1%	43.7%	65.1%	62.8%
	Mixcut	9.8%	6.5%	19.5%	34.9%	20.9%	20.5%
	NWSP	13.5%	11.2%	17.7%	21.9%	20.5%	10.7%
	SFCoT	72.1%	66.5%	97.2%	46.0%	72.1%	71.2%
	AMPN	40.5%	46.1%	**100.0%**	43.3%	37.2%	53.5%
	AFM	84.4%	84.9%	99.6%	47.1%	79.9%	87.6%
	L2T	80.6%	71.3%	98.6%	**53.5%**	81.0%	79.4%
	HTC	**87.6%**	**88.4%**	**100.0%**	47.6%	**90.7%**	**91.6%**
IncRes-v2	MIM	4.2%	3.7%	3.7%	**100.0%**	4.7%	3.5%
	DIM	5.1%	2.3%	3.3%	**100.0%**	4.2%	4.2%
	ILA	6.5%	4.2%	5.6%	87.9%	5.1%	12.6%
	SIM	6.1%	4.2%	6.1%	91.2%	4.7%	8.8%
	Admix	5.1%	4.2%	6.5%	97.7%	6.1%	9.3%
	SSA	4.7%	5.1%	5.1%	99.5%	6.5%	10.7%
	GIM	6.1%	5.6%	5.1%	99.5%	5.6%	8.8%
	BSR	9.3%	14.9%	12.1%	**100.0%**	16.3%	19.5%
	Mixcut	4.2%	3.3%	3.7%	67.4%	4.2%	8.8%
	NWSP	4.2%	3.7%	3.7%	58.1%	6.5%	6.1%
	SFCoT	7.0%	8.4%	10.7%	**100.0%**	14.4%	15.4%
	AMPN	4.2%	4.7%	5.1%	**100.0%**	4.2%	8.8%
	AFM	11.1%	11.6%	11.6%	95.1%	16.0%	15.1%
	L2T	13.4%	13.0%	14.2%	98.6%	16.5%	18.7%
	HTC	**15.1%**	**17.7%**	**14.9%**	**100.0%**	**21.3%**	**19.7%**
Inc-v3	MIM	11.2%	16.7%	12.1%	46.1%	98.1%	20.9%
	DIM	7.0%	11.6%	13.5%	5.1%	**100.0%**	14.0%
	ILA	7.9%	10.7%	8.8%	**50.7%**	40.0%	17.2%
	SIM	16.3%	13.7%	13.5%	40.5%	94.9%	24.2%
	Admix	34.9%	40.5%	41.9%	33.5%	94.4%	46.5%
	SSA	30.7%	39.5%	36.7%	35.4%	96.3%	44.2%
	GIM	9.8%	17.7%	11.6%	46.5%	95.8%	22.8%
	BSR	44.7%	46.5%	46.5%	36.7%	95.4%	50.7%
	Mixcut	25.1%	28.4%	17.2%	35.8%	62.8%	27.9%
	NWSP	13.5%	7.4%	8.8%	21.4%	38.6%	8.4%
	SFCoT	44.7%	55.8%	56.3%	34.9%	93.0%	62.8%
	AMPN	10.7%	18.1%	10.2%	44.2%	**100.0%**	27.0%
	AFM	56.4%	48.0%	63.6%	44.4%	98.7%	63.1%
	L2T	57.8%	50.0%	64.2%	44.5%	92.7%	68.4%
	HTC	**69.8%**	**68.9%**	**84.9%**	47.6%	97.3%	**87.6%**
GoogLeNet	MIM	19.1%	26.1%	28.8%	41.4%	22.8%	**100.0%**
	DIM	17.2%	28.8%	29.8%	7.0%	29.8%	**100.0%**
	ILA	13.0%	16.3%	11.6%	**53.0%**	13.5%	53.0%
	SIM	33.0%	34.4%	34.0%	41.9%	37.7%	**100.0%**
	Admix	42.8%	40.0%	46.1%	43.3%	48.4%	**100.0%**
	SSA	39.5%	38.6%	38.6%	44.6%	43.7%	99.5%
	GIM	24.2%	25.1%	26.1%	45.1%	24.7%	**100.0%**
	BSR	64.2%	57.2%	71.2%	44.2%	71.2%	95.8%
	Mixcut	18.6%	21.4%	27.9%	38.6%	34.9%	40.5%
	NWSP	12.1%	7.9%	7.4%	27.4%	16.7%	17.7%
	SFCoT	69.8%	69.8%	78.6%	40.0%	74.4%	98.1%
	AMPN	24.2%	28.3%	24.2%	40.9%	21.9%	**100.0%**
	AFM	71.3%	81.3%	87.1%	46.7%	84.9%	99.6%
	L2T	73.9%	78.4%	89.3%	46.3%	82.1%	98.6%
	HTC	**74.9%**	**83.3%**	**89.8%**	49.8%	**86.2%**	**100.0%**

3) Comparison Methods: To demonstrate the effectiveness of the proposed method, we compare HTC against 15 competitive adversarial attack methods, including MIM [14], DIM [23], ILA [43], SIM [19], Admix [75], SSA [25], AITL [34], Mixcut [27], NWSP [66], GIM [42], BSR [47], SFCoT [59], AMPN [60], AFM [61], and L2T [35].

4) General Implementation: All models are trained following the original data splits provided by each dataset. The

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TABLE III

ATTACK SUCCESS RATES ON THE SIRI-WHU DATASET AGAINST DIFFERENT MODELS. THE BEST ATTACK PERFORMANCE ON EACH MODEL IS BOLDED

Model	Attack	Vgg-16	Res-34	Dense-121	IncRes-v2	Inc-v3	GoogLeNet
Vgg-16	MIM	**100.0%**	72.2%	76.7%	37.2%	76.2%	77.4%
	DIM	**100.0%**	73.9%	76.2%	16.2%	74.1%	76.5%
	ILA	80.0%	39.8%	28.5%	44.0%	63.8%	44.0%
	SIM	99.5%	72.0%	76.5%	35.1%	79.3%	80.5%
	Admix	99.8%	74.8%	78.5%	36.4%	83.9%	83.4%
	SSA	**100.0%**	79.5%	80.0%	44.1%	82.7%	86.9%
	GIM	**100.0%**	75.5%	73.9%	40.8%	78.1%	79.1%
	BSR	**100.0%**	92.3%	92.6%	39.6%	84.2%	92.1%
	Mixcut	57.4%	24.5%	31.5%	24.5%	58.8%	43.9%
	NWSP	47.1%	17.2%	8.9%	29.2%	35.1%	25.2%
	SFCoT	**100.0%**	96.8%	93.1%	43.6%	88.9%	95.8%
	AMPN	**100.0%**	73.9%	72.2%	38.1%	78.4%	79.3%
	AFM	99.3%	90.1%	92.5%	41.9%	83.1%	96.0%
	L2T	**100.0%**	**98.6%**	98.1%	46.8%	93.2%	98.4%
	HTC	**100.0%**	**98.6%**	**99.1%**	**50.8%**	**94.1%**	**99.3%**
Res-34	MIM	72.9%	**100.0%**	56.7%	40.7%	70.6%	63.5%
	DIM	42.6%	**100.0%**	60.5%	6.6%	51.3%	59.1%
	ILA	65.4%	83.1%	30.1%	44.5%	64.5%	44.7%
	SIM	73.2%	**100.0%**	66.4%	38.8%	73.4%	70.6%
	Admix	74.8%	**100.0%**	71.8%	39.1%	80.7%	74.0%
	SSA	76.0%	**100.0%**	67.1%	41.3%	75.3%	70.8%
	GIM	74.6%	**100.0%**	54.1%	43.5%	69.9%	62.1%
	BSR	86.6%	**100.0%**	92.8%	44.1%	85.9%	91.1%
	Mixcut	56.5%	38.4%	36.5%	34.1%	58.4%	43.5%
	NWSP	32.5%	33.2%	11.1%	28.5%	33.6%	21.9%
	SFCoT	85.9%	**100.0%**	87.4%	44.1%	84.2%	93.1%
	AMPN	72.0%	98.4%	44.2%	40.2%	66.1%	55.1%
	AFM	84.9%	**100.0%**	91.1%	44.2%	84.7%	94.8%
	L2T	89.9%	**100.0%**	**93.6%**	44.6%	**91.6%**	95.3%
	HTC	**92.2%**	**100.0%**	93.4%	**48.5%**	90.1%	**96.0%**
Dense-121	MIM	71.1%	66.8%	**100.0%**	40.2%	68.0%	63.3%
	DIM	40.7%	68.0%	**100.0%**	7.3%	53.2%	54.4%
	ILA	66.1%	43.3%	57.2%	45.7%	63.5%	44.9%
	SIM	79.1%	79.3%	**100.0%**	40.2%	75.5%	79.5%
	Admix	85.6%	87.1%	**100.0%**	40.1%	80.0%	85.4%
	SSA	78.0%	76.2%	**100.0%**	42.8%	73.5%	70.1%
	GIM	72.0%	65.9%	99.8%	42.4%	62.1%	58.8%
	BSR	91.3%	94.3%	99.8%	43.6%	84.4%	90.8%
	Mixcut	31.8%	23.5%	37.9%	38.6%	51.8%	39.1%
	NWSP	30.4%	17.2%	20.5%	29.7%	33.4%	22.6%
	SFCoT	91.1%	98.3%	**100.0%**	45.3%	88.1%	97.0%
	AMPN	70.1%	59.6%	96.5%	40.0%	64.9%	56.7%
	AFM	95.4%	99.3%	**100.0%**	49.2%	88.0%	98.6%
	L2T	95.4%	98.3%	**100.0%**	51.9%	91.5%	98.3%
	HTC	**95.8%**	**100.0%**	**100.0%**	**52.1%**	**92.1%**	**99.8%**
IncRes-v2	MIM	29.4%	13.2%	10.8%	99.5%	36.7%	18.4%
	DIM	16.7%	9.2%	12.5%	99.5%	28.7%	13.2%
	ILA	40.9%	21.4%	15.8%	82.6%	41.2%	30.4%
	SIM	33.4%	18.4%	15.1%	89.9%	35.1%	23.8%
	Admix	40.1%	22.5%	20.1%	95.1%	37.1%	24.0%
	SSA	38.4%	20.3%	16.1%	98.5%	39.4%	23.8%
	GIM	34.8%	18.8%	15.5%	**99.8%**	37.9%	23.1%
	BSR	57.2%	47.0%	48.8%	**99.8%**	56.2%	41.8%
	Mixcut	30.4%	13.4%	10.4%	49.4%	34.8%	17.7%
	NWSP	26.8%	14.4%	9.2%	46.8%	28.7%	19.5%
	SFCoT	51.7%	37.4%	36.9%	**99.8%**	46.8%	33.4%
	AMPN	35.3%	15.5%	16.5%	98.8%	37.7%	21.4%
	AFM	60.9%	49.2%	52.0%	98.4%	53.9%	45.6%
	L2T	64.7%	59.3%	59.8%	99.3%	43.5%	50.8%
	HTC	**65.8%**	**61.4%**	**61.4%**	99.3%	**56.5%**	**51.7%**
Inc-v3	MIM	56.7%	30.8%	25.2%	40.2%	**100.0%**	47.5%
	DIM	20.2%	21.4%	20.9%	6.4%	**100.0%**	31.3%
	ILA	48.2%	30.4%	21.2%	43.3%	84.5%	44.2%
	SIM	64.5%	38.4%	32.7%	39.1%	97.2%	59.8%
	Admix	66.1%	44.8%	36.6%	30.9%	93.6%	48.5%
	SSA	56.2%	36.6%	24.8%	34.4%	84.4%	42.1%
	GIM	55.1%	34.4%	28.2%	42.8%	**100.0%**	47.1%
	BSR	75.3%	54.5%	50.3%	34.9%	98.8%	66.6%
	Mixcut	63.8%	47.1%	56.2%	33.2%	77.7%	49.2%
	NWSP	22.6%	12.5%	9.2%	24.2%	53.9%	16.5%
	SFCoT	71.5%	50.7%	42.8%	33.9%	99.0%	63.6%
	AMPN	55.3%	28.5%	21.4%	40.2%	94.6%	41.9%
	AFM	83.9%	79.5%	79.8%	**45.4%**	96.9%	95.8%
	L2T	82.8%	79.8%	81.7%	43.5%	99.8%	94.1%
	HTC	**84.7%**	**85.4%**	**89.4%**	40.2%	96.5%	**96.2%**
GoogLeNet	MIM	64.9%	49.7%	46.1%	40.7%	65.9%	**100.0%**
	DIM	40.5%	60.3%	54.1%	7.3%	56.5%	99.8%
	ILA	56.5%	38.6%	27.3%	**45.9%**	58.4%	68.9%
	SIM	74.8%	60.2%	57.9%	39.3%	72.5%	99.8%
	Admix	79.2%	69.1%	67.3%	39.1%	79.0%	**100.0%**
	SSA	68.8%	64.1%	52.5%	41.1%	75.0%	**100.0%**
	GIM	65.2%	51.5%	45.4%	42.8%	65.7%	99.8%
	BSR	80.5%	84.2%	85.2%	41.8%	84.2%	**100.0%**
	Mixcut	50.1%	36.2%	35.3%	39.5%	61.9%	59.3%
	NWSP	31.1%	14.8%	10.4%	27.1%	33.7%	38.4%
	SFCoT	81.9%	83.2%	80.0%	42.6%	88.6%	**100.0%**
	AMPN	65.9%	40.5%	36.7%	40.2%	61.9%	97.4%
	AFM	87.8%	91.5%	93.2%	43.8%	91.1%	99.8%
	L2T	92.2%	**94.6%**	94.6%	45.4%	93.6%	99.8%
	HTC	**93.2%**	**94.6%**	**95.3%**	**45.9%**	**94.8%**	**100.0%**

attack performance is evaluated on the subset of test images that can be correctly classified by all six DNN architectures. During iterative perturbation generation, all attack methods are implemented under identical settings, with the number of iterations T, perturbation magnitude ϵ, and step size α set to 30, 16, and 1, respectively. For the hyperparameter configuration of HTC, the action execution multiplicities at each hierarchy are set to 2, while the learning rate η, the

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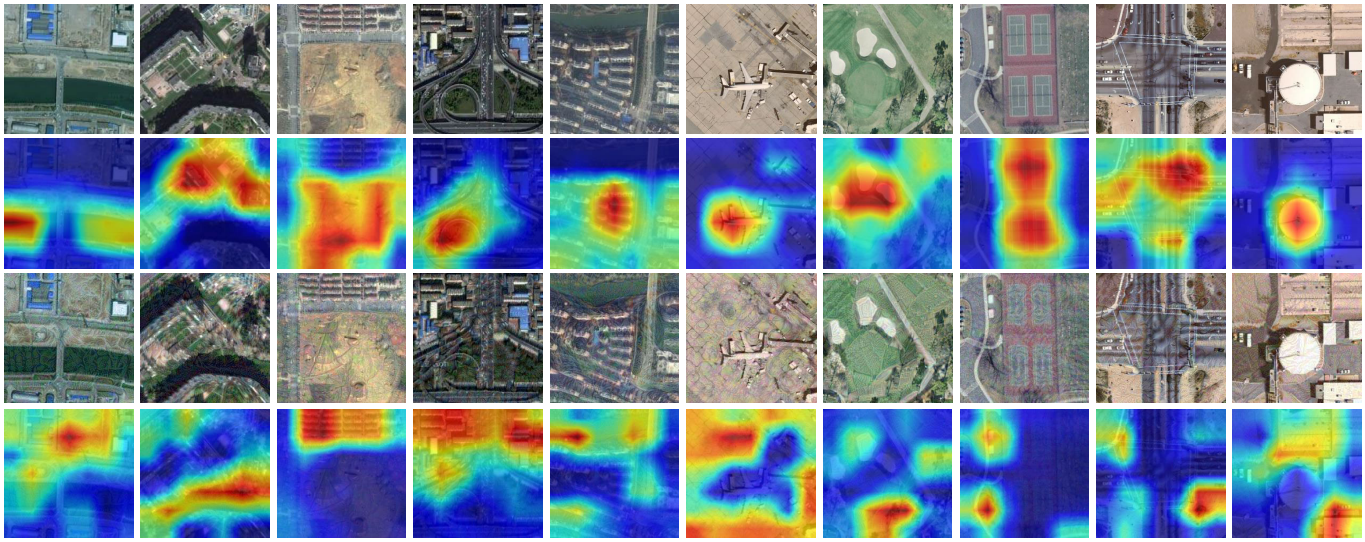


Fig. 4. Visualization of attention shift in the target model. The clean inputs and their attention maps obtained using Grad-CAM [76] are shown in the upper two rows, whereas the adversarial examples generated by HTC and their corresponding attention maps are presented in the lower rows.

clipping range ξ, and the initial weight coefficient λ for updating the policy networks are configured as 0.05, 0.2, and 0.5, respectively. To ensure a fair comparison, the transformation multiplicity used during gradient calculation is set to 8 for all methods, i.e., $K_1K_2K_3$. Following common practice in existing attack methods, both adversarial example generation and policy network optimization are performed in batches of size 4. The experiments are implemented in the PyTorch framework with NVIDIA RTX 3090 GPU support.

B. Comparison Experiments

1) Quantitative Analysis: We conduct a comprehensive comparison between HTC and other competitive attack methods across different DNN architectures. The attack performance is quantified by the attack success rate, which is defined as the percentage of adversarial examples that fool the target model into producing incorrect predictions. The results are reported in Tables II and III, where each row corresponds to a source model used for generating adversarial examples, and the associated columns correspond to the target models.

According to the results, the adversarial examples generated by HTC consistently demonstrate stronger attack performance against different models than the compared methods on each dataset. Under the white-box setting, the proposed method achieves near-perfect attack success rates, indicating that the intrinsic attack capacity is effectively preserved while enhancing transferability through input transformations composed through the multilevel policy networks. In contrast, directly imposing intermediate-layer losses on perturbation updates, e.g., ILA, induces clear degradations in attack strength against source models. Then, with respect to the transferability across different models under the black-box setting, HTC achieves a clear performance margin over the other methods in misleading unknown target models. For example, on the UCM dataset, when Inc-v3 is adopted as the source model, many attack methods attain high white-box success rates. However, when the generated adversarial examples are transferred to different target models, HTC surpasses the best-performing baseline by more than 14% on average. Moreover, we observe that the adversarial transferability associated with IncRes-v2 is generally lower than that with other DNN architectures. This mainly stems from its hybrid Inception-ResNet design, which leads to larger discrepancies in discriminative cues between IncRes-v2 and other architectures. Furthermore, SFCoT and AFM typically achieve stronger black-box attack performance than methods that rely on single-domain input transformations. The integration of spatial- and frequency-domain transformations can be regarded as the enrichment of first-hierarchy factors in our formulation. This finding further confirms our motivation in reinforcing adversarial transferability by expanding the decision dimensionality of transformation selection.

2) Qualitative Analysis: To provide an intuitive demonstration of how the proposed method achieves transferable adversarial attacks for remote sensing image classification, we further conduct a qualitative analysis in this section. Specifically, we compare the original inputs and adversarial counterparts generated using Res-34 as the source model, along with the attention maps produced by Vgg-16 when making predictions as the target model. According to the pairwise results in Fig. 4, the visual difference between the original input images and the corresponding adversarial examples is nearly indistinguishable. Despite this visual similarity, the attention of the victimized model is largely shifted from the discriminative regions associated with the underlying land cover types, such as aircraft in airport scenes, toward semantically irrelevant regions. As a result, the target model is driven to produce incorrect classification results under the black-box setting.

The strong adversarial transferability of HTC and the vulnerabilities of DNNs demonstrated above also highlight the necessity of developing effective defense mechanisms in remote sensing applications. To better resist adversarial

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TABLE IV

ATTACK SUCCESS RATES AGAINST DIFFERENT DEFENSES ON THE SIRI-WHU DATASET. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Model	Vgg-16			GoogLeNet		
Methods	AT	R&P	JPEG	AT	R&P	JPEG
DIM	20.9%	71.7%	65.4%	10.1%	50.1%	37.1%
SFCoT	46.8%	87.1%	86.8%	32.0%	86.4%	82.4%
L2T	57.4%	89.4%	88.2%	35.3%	87.8%	82.8%
HTC	**75.8%**	**90.2%**	**89.7%**	**55.1%**	**92.5%**	**89.9%**

TABLE V

ABLATION EXPERIMENTS ABOUT CONTRIBUTIONS OF EACH COMPONENT. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

CAE	PPO	HRM	Vgg-16	Res-34	Dense-121	IncRes-v2
✘	✘	✘	69.3%	72.6%	84.4%	46.1%
✔	✘	✘	70.7%	73.3%	83.1%	46.3%
✘	✔	✘	70.7%	75.8%	84.7%	45.6%
✔	✔	✘	73.5%	80.0%	87.0%	48.8%
✘	✘	✔	68.0%	74.9%	82.8%	47.4%
✘	✔	✔	71.6%	78.6%	84.9%	48.5%
✘	✔	✔	73.0%	80.5%	89.3%	48.7%
✔	✔	✔	**74.9%**	**83.3%**	**89.8%**	**49.8%**

perturbations, the effectiveness of countermeasures could potentially be improved by incorporating remote sensing-specific characteristics and auxiliary information. For AT, integrating remote sensing image characteristics into adversarial example generation can facilitate the exposure of underlying vulnerabilities, and their inclusion during training encourages the learning of more robust decision boundaries. Meanwhile, for input reconstruction, critical semantic information can be better recovered from adversarial examples through effective exploitation of auxiliary data sources and domain knowledge.

3) Attack Against Defense Strategies: In addition to normally trained models, we further evaluate the attack performance of different methods against various defense strategies. The experiments are conducted on the SIRI-WHU dataset, with Inc-v3 serving as the target model. Several defense strategies, including AT [12], R&P [77], and JPEG [55], are applied to the target model. The attack performance of adversarial examples generated on Vgg-16 and GoogLeNet is reported in Table IV. The results show that HTC consistently achieves superior adversarial transferability, further demonstrating its strong attack effectiveness across practical model deployment settings.

C. Ablation Studies

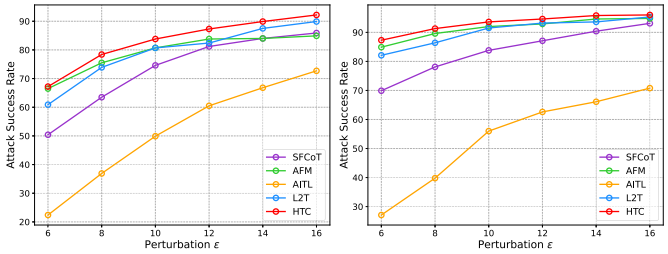
1) Contribution of Each Component: There are three main components in the proposed HTC framework: the CAE strategy, the PPO-based update of policy networks, and the HRM. In this section, we conduct ablation experiments on the UCM dataset to validate the contribution of each component to the reinforcement of adversarial transferability.

Specifically, the attack performance of adversarial examples generated on GoogLeNet against other target models under

TABLE VI

ANALYSIS OF COMPUTATIONAL COMPLEXITY. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

Methods	Time (s)	Vgg-16	Dense-121	IncRes-v2	Inc-v3
Random	901	76.3%	83.7%	47.9%	75.8%
AITL	2643	47.0%	49.8%	41.4%	44.2%
L2T	1711	79.4%	88.5%	49.1%	79.9%
CAE+PPO	955	82.3%	90.2%	50.1%	84.0%
HTC	1041	**85.6%**	**91.6%**	**50.2%**	**84.2%**



(a) (b)

Fig. 5. Attack success rates with varying perturbation magnitudes. (a) Vgg-16. (b) GoogLeNet.

TABLE VII

ATTACK SUCCESS RATE OF HTC WITH DIFFERENT EXECUTION MULTIPLICITY SETTINGS. THE BEST RESULTS ARE HIGHLIGHTED IN BOLD

K_1	K_2	K_3	Vgg-16	Dense-121	IncRes-v2	Inc-v3
1	4	2	87.1%	89.9%	43.8%	85.4%
1	2	4	89.2%	92.5%	43.8%	85.9%
2	1	4	91.5%	93.1%	48.0%	89.9%
2	4	1	91.1%	92.2%	47.1%	89.9%
2	2	2	**92.2%**	**93.4%**	**48.5%**	**90.1%**

different component configurations is reported in Table V. We adopt randomly selecting transformations from a given scope for gradient calculation as the baseline. First, simply expanding the range of available input transformations by randomly composing factors from different hierarchies leads to a certain degree of improvement even though the selection process remains nonlearnable. This indicates the effectiveness of hierarchical transformation factorization in diversifying input pattern simulation. Nevertheless, the improvement achieved by merely increasing transformation diversity through this nonlearnable composition strategy remains modest. Then, according to the results in the third and fourth rows, the use of PPO consistently leads to improved attack performance under both the baseline and hierarchical settings. This demonstrates its effectiveness in adapting the transformation policies to the evolving gradient landscape, thereby enabling more generalizable perturbation updates across different model architectures. Moreover, directly imposing integrated loss information on gradient manipulation may even degrade attack performance compared with the baseline. In contrast, repurposing such hybrid supervision signals to guide the optimization of policy networks can effectively improve transferability without sacrificing intrinsic attack capacity. Finally, combining all three components into the proposed HTC framework achieves

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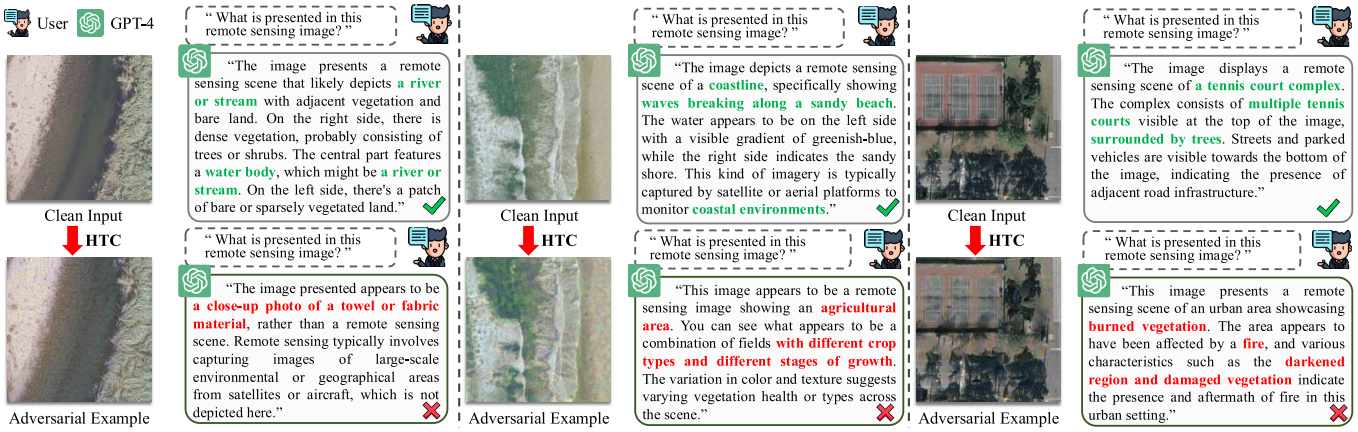
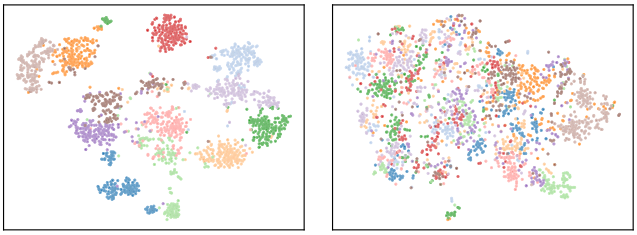
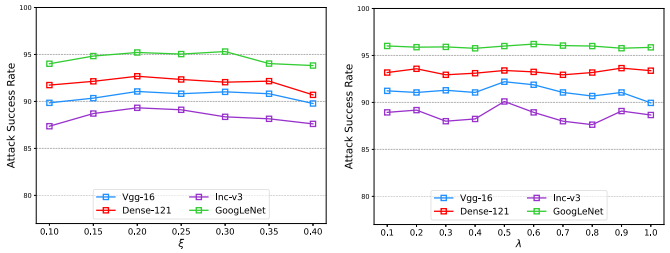


Fig. 6. Visualization of original clean inputs and adversarial examples generated by HTC against GPT-4. Correct and false descriptions of the content presented in each image are highlighted in green and red, respectively.



(a) (b)

Fig. 7. Visualization of representations for clean inputs and HTC-generated adversarial examples. Different classes are indicated by different colors. (a) Clean. (b) HTC.



(a) (b)

Fig. 8. Attack success rate of HTC with different settings of (a) ξ and (b) λ.

the best overall attack performance, confirming their complementary contributions to addressing transferability challenges arising from remote sensing image characteristics. In addition, we evaluate a variant where the composed transformations are applied independently during gradient calculation. As shown in the second-to-last row, this setting leads to a performance drop compared with the full model, which confirms the effectiveness of the transformation ensemble grouping strategy.

2) Computational Complexity: In addition to the quantitative and qualitative evaluations of the attack performance, we also analyze the computational efficiency of HTC in this section. The experiments are conducted on the UCM dataset using Res-34 as the source model. The total time required by each method to generate adversarial examples, along with the corresponding attack success rates on different target models, is reported in Table VI. From the results, HTC incurs only a marginal computational overhead compared with randomly selecting transformations from a fixed scope. Moreover, HTC achieves higher attack performance while requiring noticeably less computational time than other transformation ensemble methods. In particular, we further analyze the cost of realizing the learnable transformation composition via RL optimization (CAE+PPO), which increases computation by only about 6% over the random-selection baseline.

3) Performance With Different Perturbation Magnitudes: As a critical factor in the generation of adversarial examples, the attack success rate is closely related to the perturbation magnitude. To further demonstrate the effectiveness of the proposed method, we compare HTC with two leading adversarial attack methods for remote sensing image classification (SFCoT and AFM), as well as the latest learnable transformation-ensemble methods (AITL and L2T) across different perturbation magnitudes. Specifically, the experiments are conducted on the SIRI-WHU dataset by adopting Res-34 as the source model to generate adversarial examples. The attack success rates against Vgg-16 and GoogLeNet with perturbation magnitudes ranging from 6 to 16 are reported in Fig. 5. According to the results, HTC consistently achieves superior attack performance compared with other methods across different perturbation magnitudes, further demonstrating the effectiveness of the proposed method in enhancing adversarial transferability under diverse settings.

4) Extended Evaluation on LVLMs: Beyond evaluations on commonly adopted DNN architectures for remote sensing image classification, we further explore the effectiveness of the proposed attack method on large vision–language models (LVLMs) [78], [79], [80]. Specifically, ViT-B [81] is adopted as the source model to generate adversarial examples, and both the clean inputs and their adversarial counterparts are subsequently fed into the GPT-4 [82] API to assess whether HTC can induce incorrect understanding of image content under the black-box setting. As illustrated in Fig. 6, although visually imperceptible and not explicitly designed for visual question answering tasks, HTC is capable of fooling GPT-4

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into producing erroneous descriptions that diverge markedly from the true scene content.

Apart from the attack on LVLM-based image content understanding, we also examine the impact of HTC on the visual representations learned by pretrained vision–language models, whose discriminability is crucial for many important unsupervised remote sensing tasks, such as clustering [83], [84] and retrieval [85]. Specifically, we visualize the representations extracted by RemoteCLIP [86] with a ViT-L/14 backbone for clean images from the SIRI-WHU dataset and their adversarial counterparts generated by HTC. According to the t-SNE [87] visualization in Fig. 7, the representations of clean images exhibit clear semantic consistency, whereas HTC can significantly reduce the interclass separability and intraclass compactness of the representations.

5) Parameter Analysis: In this section, we analyze the influence of parameter settings involved in the HTC process, including the clipping range ξ in the surrogate objective, the initial weight coefficient λ in the HRM, and the execution multiplicity $\{K_1, K_2, K_3\}$ at different hierarchies. The experiments are conducted on the SIRI-WHU dataset using Res-34 as the source model.

First, we set λ to 0, i.e., updating the policy networks based on only the output-layer loss information, and vary ξ from 0.1 to 0.4. According to the results shown in Fig. 8(a), the attack success rates against different target models improve as ξ increases within a certain range but tend to degrade when ξ becomes excessively large. This is because a too-small clipping range can overly restrict the policy updates, whereas an excessively large range undermines the stability constraint of PPO and may cause training collapse. Next, we fix ξ to 0.2 and evaluate the attack performance when updating policy networks by the HRM, with λ varied from 0.1 to 1. From Fig. 8(b), we observe that the attack performance remains stable across different values of λ. Based on the above analysis, we set $\xi = 0.2$ and $\lambda = 0.5$ in HTC and further investigate the impact of execution multiplicity allocation across different hierarchies. The attack performance of HTC under different settings of $\{K_1, K_2, K_3\}$ is reported in Table VII, where the total transformation multiplicity is set to 8. The results indicate that uniformly assigning the execution multiplicity of 2 to each hierarchy yields relatively better attack performance.

V. Conclusion

In this article, we have proposed HTC, a CAE framework to reinforce the adversarial transferability for remote sensing image classification. Different from existing attack methods, HTC enriches the diversity of simulated input patterns by adaptively optimizing transformation composition policies. Specifically, hierarchical transformation factors are progressively determined by multilevel policy networks to capture transferable gradient cues. Then, the multilevel policy networks are iteratively updated in an actor–critic manner to continuously adapt to the evolving gradient landscape. Moreover, an HRM is introduced to provide a more informative action value for policy optimization, achieving further transferability enhancement while maintaining intrinsic attack capacity. In future work, we plan to further investigate adversarial attacks and defenses in other remote sensing tasks, such as target detection and semantic segmentation. In addition, given the growing adoption of LVLMs in the remote sensing domain, we will further investigate their adversarial robustness in terms of both content understanding and energy efficiency, thereby facilitating the reliable deployment of deep learning models in real-world remote sensing applications.

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